

Shabbir Hathi

AI Engineer & Backend Developer

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PROFESSIONAL SUMMARY

AI and backend engineer with 4 years shipping production LLM systems, multi-agent pipelines, and SaaS platforms end to end. Architected a 16-agent pipeline that turns a plain-language prompt into a playable browser learning game, cut an AI interview agent's per-turn latency by ~60% and its prompt tokens by ~47%, and built the analytics platform whose desktop agents run ~9 hours a day on 60+ machines. Clients come back: one first engaged me in 2024 and returned in 2026 to rebuild a second product.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

AI & ML: RAG, LangChain, LangGraph, multi-agent orchestration, prompt engineering, hybrid/RRF retrieval, embeddings, Anthropic Claude, OpenAI, Whisper, Perplexity API, Groq, Stable Diffusion, PyTorch, TensorFlow, scikit-learn, OpenCV, YOLO

Backend & Frontend: FastAPI, Django/DRF, Flask, NestJS, Celery, Socket.IO, SSE, REST, OpenAPI, JWT, SQLAlchemy, Prisma, Alembic, React, Next.js, TanStack Query, Vite, TailwindCSS, PyQt6

Data & Infrastructure: PostgreSQL, MySQL, Redis, Qdrant, ChromaDB, pgvector, Supabase, Pandas, NumPy, AWS (EC2, S3, SQS, Bedrock), Docker, Nginx, GitHub Actions, Playwright

Integrations: Stripe, RevenueCat, SCORM 1.2, LTI 1.3, PostHog, SerpAPI, Gamma AI, ElevenLabs, OAuth, magic-link auth, web scraping (BeautifulSoup, Selenium)

EXPERIENCE

iRoid Solutions

Nov 2024 - Present

AI Engineer & Backend Developer · Part-time

Surat, India

- Architected and solo-built Learning Games Studio for agency client Strive Studio: 16 specialised AI agents in a PostgreSQL checkpointed TypeScript state machine that turn a plain-language prompt into a complete, playable browser learning game - ~15,000 LOC, a 21-path OpenAPI spec, and SCORM 1.2 / LTI 1.3 LMS delivery.
- Built its self-healing QA chain (syntax gate -> LLM repair -> completeness audit -> Playwright and vision-model visual QA -> runtime auto-debugger) and a runtime-editable model registry metering 100% of LLM calls, with measured build cost from \$0.10 (Rapid tier) to \$1.28 (Advanced).
- Designed, shipped, and operate iRoid Tracker, a 4-codebase time-tracking and analytics platform whose native Windows and macOS PyQt6 agents run ~9 hours a day on 60+ developer machines, streaming to a FastAPI/MySQL backend and React dashboard across ~40,000 LOC and 69 REST endpoints.
- Ran a 34-item stability audit on Tracker and shipped 30 of them: halved server load average (2.09 -> 0.48) while adding Celery and Redis, cut heartbeat traffic to a third, reduced screenshot storage 80-90%, and eliminated timer drift that had under-counted up to 1h 47m/day per employee.
- Managed and built KEECZ for a German food-supply-chain client during a two-week embedded on-site sprint: a 39-endpoint Django/DRF API with a deterministic 260-point sustainability scoring engine, a 1,014-article product catalog, dual-provider LLM PDF extraction, and an SSE-streamed AI chatbot.
- Built a production Flask RAG backend (Qdrant + LangChain) with multi-provider LLM routing across OpenAI, Mistral, Cohere, and AWS Bedrock - 35% lower chat latency and 22% lower token cost - plus a 10-tool LangChain event-planning agent (Wenue) and a self-hosted vector database API (<200ms p95, >1M chunks indexed).

Tech: Python, FastAPI, Django/DRF, Flask, TypeScript, NestJS, Next.js, React, LangChain, Anthropic Claude, OpenAI, Qdrant, MySQL, PostgreSQL, Celery, Redis, PyQt6, Docker, Nginx, AWS, GitHub Actions

Differenz System

Sep 2022 - May 2024

AI Engineer & Backend Developer

Surat, India

- Built an AI dog breed identifier (Django REST + YOLO detection + custom-trained classifiers) covering 170+ breeds at 95%+ accuracy - 10K+ Play Store downloads at a 4.6-star rating across 243 reviews.
- Fine-tuned Stable Diffusion for textile pattern generation with PSD colour-separable layer export, post-generation recolouring APIs, and garment try-on previews, served over a Celery/Redis GPU job queue.
- Built a Django recommendation API (cosine similarity plus Apriori association-rule mining) and an automated ESPN NBA/NCAA scraping pipeline (BeautifulSoup/Selenium into MySQL, cron-scheduled with CSV backups).

Tech: Python, Django, MySQL, PyTorch, TensorFlow, OpenCV, YOLO, Stable Diffusion, CUDA, Celery, Redis, BeautifulSoup, Selenium, Pandas, scikit-learn

FREELANCE & CONTRACT EXPERIENCE

Unifeyn

Jun 2026 - Aug 2026

Lead Engineer & Product Strategist

Remote

- Took over an unstable, vibe-coded production AI study app and rebuilt it from scratch across 20+ migrations, moving every existing user, workspace, and RAG document with zero data loss via a purpose-built Node.js export / re-embed / idempotent-SQL migration tool.
- Designed the production PostgreSQL + pgvector schema (workspaces, membership roles, knowledge resources, chunks/embeddings) with RLS-based Owner/Editor/Viewer collaboration, replacing a flat, owner-only data model, and shipped invites, share links, referral, and affiliate programs.
- Root-caused a RevenueCat webhook mapping bug that silently wrote an invalid plan tier and left 100% of paying weekly subscribers capped at free-tier limits; separately restored self-serve cancellation by tracing it to a blank Stripe Customer Portal configuration.
- Designed a 6-tier credit subscription system metered on real per-model API cost and validated it against an App-Store-vs-Stripe margin model; authored the competitive gap analysis and 12-strategy growth playbook that set the rebuild's roadmap.
- Repeat engagement: the same client first hired me in 2024 to build SupportJoy.

Tech: React, TypeScript, TanStack Start, Supabase, PostgreSQL, pgvector, Deno Edge Functions, OpenAI, Whisper, ElevenLabs, RevenueCat, Stripe, PostHog

Wizly AI

Dec 2025 - Mar 2026

AI Engineer

Remote (Singapore / US distributed team)

- Sole owner of the AI and agent layer on a GTM decision-intelligence SaaS platform, shipping 10+ features across ingestion, retrieval, agentic interviews, and generative reporting.
- Architected the Expert Interview agent on LangGraph: it verifies each expert claim against a date-aware Perplexity web search, scores responses across 8 quality dimensions plus a cumulative depth score, and applies 4 exit conditions to either close the interview or generate a targeted follow-up question.
- Cut per-turn latency from ~20-30s to ~8-10s by caching per-request DB context and parallelizing three validation/scoring calls onto a Haiku-based async pipeline, and reduced prompt token usage by ~47% using last-exchange-only context windows.
- Built hybrid RRF (dense + sparse) retrieval with an LLM relevance filter for the expert search API, and delivered 3 Gamma-AI-backed endpoints generating McKinsey/BCG-style PDF and PPT board reports.

Tech: Python, FastAPI, LangChain, LangGraph, Anthropic Claude, OpenAI, Perplexity API, Groq, Qdrant, PostgreSQL, AWS S3, AWS SQS, Celery, Gamma AI, CopilotKit

PriveGuard

Aug 2024 - Dec 2024

Full-Stack Developer

Remote

- Built an anti-piracy SaaS for digital creators: SerpAPI-driven automated detection, a domain whitelisting system to exclude legitimate sources, Stripe subscription billing, a customer dashboard, and monthly analytics reporting.
- Shipped a separate Streamlit/Flask internal admin application for detection review and asset management, and delivered several React + Django product websites on a shared component and REST architecture.

Tech: Django, Python, Bootstrap, Jinja2, Stripe API, SerpAPI, Streamlit, Flask, React

SupportJoy

Jun 2024 - Oct 2024

Full-Stack Developer

Remote

- Built an AI chatbot SaaS letting businesses train an agent on their own content: RAG over uploaded PDF/TXT/DOC/PPT files and scraped URLs, an embeddable JavaScript widget, magic-link authentication, Stripe subscriptions, and per-account token metering.
- Delivered the full product solo, from dashboard and ingestion through billing and the embed. The client returned in 2026 to have me rebuild a second product, Unifeyn.

Tech: Django, Python, JavaScript, Bootstrap, Stripe API, RAG, embeddings, web scraping

EDUCATION

Bachelor of Engineering, Information Technology

May 2022

Gujarat Technological University

Gujarat, India